Using the ARMADA Algorithm to Analyze Physiological Marker Patterns in Presumed Emotional States

Abstract

Affective computing bridges the gap between human emotions and machine responsiveness through computational modeling. This study investigates detecting sequential patterns of physiological transitions during emotional episodes using the ARMADA algorithm. By analyzing datasets with synchronized cardiovascular, electrodermal, and skin temperature features, the study validated ARMADA's effectiveness for temporal pattern discovery. The generated association rules were proven statistically significant via permutation tests, with over 50 rules meeting rigorous cross-dataset validation criteria. Quantitative analysis revealed significant findings regarding the predominance of moderate affective states, HRV synchronicity, and psychological lag in transition logic.

Keywords: Physiological Rule Mining, Emotion Recognition, ARMADA Algorithm, Affective Computing, Cross-Dataset Analysis

1. Introduction

Automatic emotion recognition plays a crucial role in Affective Computing, offering the potential for continuous assessment of human emotional states [**?**]. Emotion recognition systems are built on diverse modalities, most notably facial expressions. Still, researchers have long also recognized the potential of physiological signals for emotion assessment. This stems from the fact that unlike facial expressions or voice, physiological responses such as Electrodermal Activity (EDA), Photoplethysmography (PPG), and skin temperature (SKT) are largely controlled by the autonomic nervous system and reflect genuine emotional states without conscious distortion [**?**]. In recent years, an extensive body of literature has adopted machine learning and deep learning techniques, including Support Vector Machines (SVMs), Convolutional Neural Networks (CNNs), and Recurrent Neural Networks such as LSTMs, to process these complex time-series signals [**?**]. Although these models often achieve strong classification performance, their predictions are typically difficult to interpret in mechanistic terms, making it challenging to explicitly explain how temporally evolving physiological changes relate to the emergence of a particular emotional state.

From another perspective, the past few years have highlighted the necessity of shifting emotion recognition systems away from 'black box' models toward architectures that enable explainability of their outputs [**?**, **?**]. Despite the abundance of modern Explainable AI (XAI) techniques, rule-based systems have seen a resurgence in importance due to their naturally transparent and self-explanatory nature [**?**, **?**]. The application of sequential rule mining to physiological signals, all the while, remains in its infancy. Even though human physiological responses naturally unfold over time, very few studies actually use sequential rules to analyze them [**?**]. To address this gap the research objective is to **explore the feasibility of discovering temporal patterns within specific physiological signals, leveraging continuously annotated data through the application of the Association Rule Mining And Discovery Algorithm (ARMADA)** [**?**].

2. Research Questions

To provide a structured investigation to achieve the primary objective of this study, the two research questions were defined:

RQ1: Technical Feasibility and Rule Stability *To what extent can the ARMADA algorithm effectively extract stable and statistically significant temporal association rules from multimodal, continuously annotated physiological datasets?*,

RQ2: Pattern Discovery and Physiological Correlation *Which specific temporal patterns in physiological signals (such as HRV and EDA) are most frequently associated with transitions in emotional states according to the discovered rules?* and

RQ3: Multimodality *How does the performance and interpretability of multi-modal association rules compare to single-modal rules in describing emotional transitions?*

Due to the complexity of the first research question, the supportive sub-questions were defined and investigated: RQ1.1 (Annotation Source) *How does the source of emotional labels—specifically self-reports versus expert external annotations—influence the quality and consistency of the discovered temporal rules?* and RQ1.2 (Emotional Model) *To what extent does the choice of emotional model (discrete categories [**?**] vs. multidimensional valence-arousal space [**?**]) affect the algorithm's ability to identify meaningful patterns in physiological signals?*

3. Related Work

The presented study is aligned with the broader trend in affective computing of integrating XAI to enhance model transparency. Among the leading approaches in this field are Attention-Based, Generative-Based, and Graph-Based XAI methods [**?**]. The body of research encompasses both single-modality analyses [**?**, **?**] and multimodal frameworks [**?**].

In relation to the subject of this work, the second important branch consists of studies that map multimodal physiological manifestations to underlying cognitive processes. Among these, the most recent study by Machhi, Vilas et al. (2026) [**?**] is of particular importance. In this context, there are also studies searching for rules describing the influence of the single modalities on human affect, e.g. [**?**].

However, the studies most closely related to this work are those addressing the application of sequential pattern mining to emotion recognition. Still, these solutions have been mostly restricted to modalities other than biosignals. A significant amount of research has focused on extracting sequential emotion-aware rules from text data, particularly social media content and micro-blogs [**?**, **?**]. Similarly, sequential pattern mining has been utilized to analyze multimodal combinations of verbal and facial clues in video data, unearthing temporal patterns during interpersonal interactions [**?**]. Despite these advancements across text and video modalities, physiological data remains notably excluded from sequential rule-mining approaches, leaving an unaddressed gap regarding the explicit, universally interpretable temporal rules defining human physiological reactions.

4. Methodology

This section outlines the research framework, starting with the fundamental assumptions that guide the analysis of physiological responses to emotional stimuli as well as core methodology.

4.1. Preliminary Assumptions

To ensure the clarity and focus of the investigation, the primary assumptions and technical choices have been established.

Selection of Physiological Modalities: The study focuses on three primary types of signals: Blood Volume Pulse (BVP), EDA, and SKT. These modalities were selected for two main reasons. First, they are highly sensitive to autonomic nervous system changes, providing reliable markers for emotional arousal (EDA [**?**, **?**], BVP [**?**, **?**, **?**] and SKT [**?**]) and valence (primarily Heart Rate Variability (HRV) [**?**, **?**, **?**]). Second, they represent the most prevalent biosignals in publicly available affective computing datasets, ensuring that the proposed framework is compatible with a wide range of existing research benchmarks and allows for cross-dataset validation of the discovered rules.

Pattern Discovery Paradigm: The research framework is based on the Sequential Association Rules (SAR) mining approach. In contrast to static association rules, SAR accounts for the temporal order of events, which is essential for capturing the dynamic nature of emotional transitions and the inherent physiological lag between a stimulus and the body's response.

Algorithmic Framework: The pattern discovery process utilizes the ARMADA algorithm, engineered to extract association rules from multi-modal time-series data. Instead of raw continuous points, it relies on discretized symbolic intervals, enabling the integration of heterogeneous signals (e.g., high-frequency cardiac data and slow-varying EDA) into a unified event stream. By representing physiological changes as time-stamped symbols, ARMADA identifies complex temporal predicates—such as overlaps, contains, or starts—that extend beyond simple co-occurrence. This is essential for modeling the physiological lag between emotional stimuli and autonomic responses. Furthermore, the algorithm avoids rigid a priori structures, detecting both acute spikes and long-term trends. The resulting human-readable "If-Then" rules ensure transparency, aligning with the XAI paradigm by providing causal-like justifications for affective state classifications.

Rule Evaluation Metrics: To assess the quality and reliability of the discovered patterns, the research framework relies on standard association rule metrics—primarily *Support*, *Confidence*, and *Lift*—specifically adapted for temporal sequences.

4.2. Core Methodology

The six main steps are defined within the Core Methodology. The first one, **Data Acquisition**, involves selecting datasets that address the research questions—specifically those providing continuous annotations and multimodal biosignals. Additionally, it is required that some datasets contain self-annotations, whereas others provide external ones. A specific emotional model is not mandatory, as it was assumed that discrete models could be transformed into multidimensional ones and vice versa. Incorporating that requirement carried a risk with respect to the availability of the dataset; however, the study requires at least two externally annotated sets and two self-annotated sets. In the **Data Preprocessing and Feature Extraction** step, physiological signals are processed to extract key features, which are then subjected to normalization and discretization. This stage also involves the harmonization of labeling across disparate datasets. Depending on the specific emotional model employed, appropriate transformations are applied to unify the emotion representation. A key component of this research involves the **Experimental Design and Validation Strategy**, which establishes the protocols required to investigate the research questions. In the subsequent **Experiments Execution** phase, these designs are implemented and validated. Finally, the resulting data are analyzed to evaluate their alignment with the study's original goal and research questions (**Analysis and Evaluation**).

5. Data Acquisition

Searching for datasets combining multimodal physiological recordings with relatively continuous or densely sampled emotions annotations was conducted in an unstructured study, which encompassed analysis of the Systematic Literature Review (SLR) [**?**, **?**, **?**], interviews with other Affective Computing researchers, and direct efforts to obtain dataset access, including contacting dataset authors with formal requests, completing User Agreement Forms when required, and, in the case of the EMBOA dataset, reaching out to the person responsible for the project. As a result of this study, five multimodal emotion datasets were chosen: K-emoCon [**?**], Continuously Annotated Signals of Emotion (CASE) [**?**], Continuous Physiological and Behavioral Emotion Annotation Dataset for 360°Videos (CEAP-360VR) (hereinafter referred to as CEAP) [**?**], EMBOA [**?**], and EmoWorker [**?**]. These datasets differ in both acquisition setting and annotation protocol, covering controlled laboratory recordings as well as conversational and im-

mersive VR scenarios. They also include both self-reported annotations and externally assigned labels, which makes them suitable for evaluating rule mining under heterogeneous emotion-labeling schemes.

More specifically, EMBOA was collected in a human-robot interaction setting and combines Empatica E4 peripheral signals with externally assigned behavioral annotations (done with variable frequency), whereas K-emoCon captures dyadic conversations and provides self, partner, and external affective ratings (done every 5 seconds) alongside physiological recordings. CASE represents a controlled laboratory dataset with high-resolution physiological signals and continuous self-assessment gathered using a joystick interface (with frequency 20 Hz), while CEAP extends this paradigm to immersive 360° audiovisual stimulation in virtual reality [**?**] (with annotation frequency equals to 10 Hz). EmoWorker, in turn, provides multimodal physiological recordings acquired in a workplace-related context together with continuous affect annotations [**?**] with a frequency ranging from 5 to 10 Hz. In each dataset (except EMBOA), the number of participants was approximately 30 (+/-2). For the EMBOA dataset, the number of participants was 16. The datasets further vary in the physiological modalities and sampling rates they provide. In addition to the physiological signals selected for this study, some datasets also incorporate electrocardiogram, electroencephalogram, respiration, eye tracking, or motion-related signals [**?**, **?**, **?**, **?**]. For all datasets (except CASE) the frequency of BVP is 64Hz, and 4Hz for EDA and SKT. The frequency of all three signals for CASE is 1000Hz.

6. Data Preprocessing and Feature Extraction

As it was already stated in Section **??**, the datasets contain either self-annotations or external ones. In the case of the K-emoCon dataset, it contains both types of annotations, therefore to ensure unambiguous characteristics (only one type of annotation) for each dataset, it is divided into two separate ones - one set for self-annotations and one for external ones (hereinafter referred to as K-emoCon and K-emoCon (ext), respectively). As a result, the input for further processing is actually 6 datasets.

Physiological features were extracted directly from the raw signals of all datasets using the same NeuroKit2-based processing pipeline [**?**], ensuring a standardized representation across modalities and datasets before temporal segmentation. HR-based features were used to characterize both the instantaneous cardiac state and short-term HRV. In particular, HR and IBI represented the current dynamics of heart activity, whereas `hrv_rmssd`, `hrv_pnn50`, `hrv_sdnn`, `hrv_cvnn`, and `hrv_hfn` quantified variations between successive beats and reflected autonomic cardiovascular regulation. EDA features were included to capture phasic electrodermal responses, with `eda_scr_amp`, `eda_scr_auc`, and `eda_peaks` describing the magnitude and occurrence of skin conductance reactions. Temperature features comprised `temp_mean` and `temp_slope`, representing the average thermal level within a segment and its temporal trend, respectively.

To synchronize disparate sampling rates across multiple datasets, continuous physiological signals and ground truth annotations were segmented using a static sliding window mechanism. A multi-resolution methodology tailored to modality reactivity parameters was applied; dynamic signals were extracted utilizing 5-second sliding windows [**?**]. Conversely, parameters inherently reliant on longer macroscopic measurement phases, specifically HRV and SKT, were captured via independent 30-second windows [**?**]. After this process, the combined dataset from all 6 datasets consists of 176 sets (one for each participant and a few additional ones due to more than one recording for few participants in EMBOA dataset) with the average rounded number of timestamped events equal to 2485 (CASE), 85 (CEAP), 524 (EMBOA), 1233 (EmoWorker), 667 (K-emoCon), and 672 (K-emoCon (ext)).

Because ARMADA requires symbolic input, continuous physiological signals were discretized into three states: Low (L), Medium (M), and High (H). To reduce subject-specific bias,

since physiological baselines and response ranges can differ markedly across individuals [**?**, **?**], we first applied participant-wise min-max normalization and then defined L/M/H using global terciles computed from the pooled normalized data. For datasets featuring native dimensional annotations (e.g., 1–9 Self-Assessment Manikin scales), arousal and valence values were normalized to a $[0, 1]$ range per dataset and divided into strict terciles: low ($[0, 0.33)$), medium ($[0.33, 0.66)$), and high ($[0.66, 1.0]$). This translation successfully yielded 1489 valid valence timestamped events and 952 unified arousal timestamped events. In the combined dataset, it resulted in 7265 events for valence (*low* - 1111, *medium* - 4728, *high* - 1426) and 12319 events for arousal (*low* - 1281, *medium* - 4546, *high* - 1502) suitable for ARMADA processing.

To enable dimensional vs. discrete comparisons, all dimensional data were mapped into a 3×3 grid of 9 categorical emotions via Russell's circumplex model. Combinations of arousal and valence terciles defined unique discrete states (e.g., *high* arousal and valence yielded *excited*; *medium* states yielded *neutral*) [**?**]. This synchronized all datasets—both inherently dimensional and dimensional-mapped discrete cohorts (EMBOA)—into a single 9-label framework. Across all datasets, this produced 4073 *neutral* tuples, followed by *calm* (947), *content* (938), *alert* (603), *stressed* (603), *upset* (553), *excited* (484), *sad* (304), and *relaxed* (207).

To enable strict association rule mining, annotation streams recorded at varied frequencies or by multiple observers were aggregated into a singular ground truth label per temporal window. For self-reported datasets (e.g., CASE, K-emoCon), high-frequency or asynchronous ratings within a physiological window were simply averaged (linear mean) to yield one synchronized dimensional label. For externally annotated datasets featuring multiple observers, independent ratings were fused into a consensus sequence before window aggregation. For K-emoCon (ext), we used the dataset's native *aggregated_external_annotations*, where up to five continuous rating streams were aggregated via majority voting (see Code Availability). In EMBOA, the native percentage agreement across multiple BORIS raters served as the proportional weight mapping categorical emotions onto Russell's dimensional space. This ensured deterministic emotion tracking across all cohorts.

Corrupted intervals and missing values were inherently filtered or marked natively by the original dataset providers, allowing our methodology to bypass artificial interpolation entirely.

7. Experimental Design

To achieve the answers to the research questions, the two experiments were defined: Experiment 1 for RQ1, and Experiment 2 for RQ2 and RQ3.

The experimental pipeline for **Experiment 1** follows a two-tier approach. Initially, temporal association rules are extracted independently for each of the six datasets to capture site-specific patterns. To evaluate the cross-dataset robustness of the ARMADA algorithm, a consensus-based validation framework is applied. By identifying the common temporal rules across six datasets, the focus is set on patterns that achieve a high level of agreement (consensus) between different experimental setups (specific for each dataset). The resulting discrete sequential rule sets were evaluated using dynamic intersections—gradually escalating from discovering generic rules shared between any two datasets, up to generating rules for the combined datasets being the union of two, three, four, or five datasets (intersecting the rules with the one not included in the combined dataset), utilizing leave-one-out apporach ("one" meaning the whole dataset). To evaluate generic cross-dataset feasibility independently of rigid dataset-specific constraints, a universal minimum support parameter of 30% (*Minsup = 0.3*) is strictly applied across this foundational phase. This threshold is set due to the assumption that mining patterns in biosignals belongs to the class of mining rare events.

The setup supports two iterations, which vary with the emotion models applied. The first one utilizes the valence-arousal model, and the second one, the uniform discrete 9-label framework.

The datasets with self-reported annotations are strictly isolated from external annotations to

prevent structural overlap. To quantify the overlap between the rule sets extracted from the self- and externally-annotated, the Jaccard Similarity Coefficient is employed. This metric provides a normalized measure of the intersection between two sets of rules, accounting for variations in the total number of rules discovered across datasets. This analysis is done only for the valence-arousal model, as it is originally used in most of the incorporated datasets.

Experiment 2 dedicated to RQ2 is built upon the Experiment 1. Building on the results of the initial phase, Experiment 2 is designed to evaluate and interpret the outcomes of Experiment 1 in a broader context. Rather than conducting a separate procedural study, this phase focuses on a deep qualitative analysis of the most reliable association rules identified previously. These high-confidence rules are scrutinized for their functional significance and their practical implications for emotion recognition within the field of affective computing. Furthermore, to address RQ3, the analysis specifically investigates the distribution and roles of the physiological modalities involved. No dedicated experiment is designed for RQ3.

In addition to the consensus-based validation framework, permutation tests were conducted to rigorously assess the statistical significance of the discovered patterns. For the first experiment, these tests were utilized to evaluate the magnitude of rule reduction, ensuring that the filtering process effectively isolates non-stochastic associations. Regarding the second and third experiments, the permutation approach served to verify whether significant rules—previously identified through performance metrics—could still be detected within shuffled datasets. This contrastive analysis demonstrates that the validated rules are not artifacts of random data arrangements but represent genuine physiological dependencies.

8. Execution of Experiments

To execute the experiments, a proprietary implementation of the ARMADA algorithm was developed in Python. The experiments were conducted on a MacBook Pro equipped with an Apple M1 chip and 16 GB of unified memory. All experiments presented in this section were conducted, and the reported results were obtained, using scripts available in the public repository [**?**]. The repository contains the implementation necessary to reproduce the experimental procedure and the obtained outcomes.

To assess the technical feasibility of temporal rule extraction from physiological, the ARMADA algorithm was applied to all six datasets under identical strict parameters: $minsup = 0.3$, $minconf = 0.5$, $maxgap = 5$ s, and $max_pattern_size = 4$. These settings universally retain only temporal rules that span up to 4 events, constrain events within a 5-second maximum gap, and appear in at least 30% of the participants with at least 50% confidence. The semantic filters were applied to ensure cross-modal relevance: rules containing only BVP/HRV/HR-related or only EDA-related states were rejected, as were rules linking exclusively physiological modalities without any emotional dimension (arousal or valence), and rules involving only a single feature type.

To investigate the influence of annotation source on rule quality, the rules extracted from all four self-annotated datasets were pooled (union) and compared against the pooled rules from both externally annotated datasets.

9. Results of Experiments

As stated in the experimental design, the base model for experiments is a multidimensional one, so all results are discussed with reference to it unless otherwise noted.

Experiment 1. ARMADA extracted between 929 and 3032 semantically filtered cross-modal rules per dataset (see Table **??** for details), capturing diverse temporal relations (*meets*, *before*, *starts*, *equals*) between physiological features and emotional dimensions. The permutation tests, involving the randomization of event sequences, resulted in a substantial rule reduc-

Table 1. Metrics for filtered rules extracted with ARMADA per dataset.

	CASE	K-emoCon	CEAP	EmoWorker	K-emoCon (ext)	EMBOA
Dimension Model	2432	1208	3032	929	1633	2972
Discrete Model	643	144	485	171	324	1184
Dimension Model (shuffled)	82	44	498	50	114	808
Reduction	96,6%	96,4%	83,6%	94,6%	93,0%	72,8%
Discrete Model (shuffled)	38	23	91	20	34	492
Reduction	94.1%	84.0%	81.2%	88.3%	89.5%	58.4%
Ratio	0.26	0.12	0.16	0.18	0.20	0.40

tion ranging from 72% to 97% across all datasets; this underscores the fact that the vast majority of patterns discovered in the original data are not mere products of event frequency, but are fundamentally dependent on the specific temporal order and causal structure of the physiological signals.

Intersection analyses revealed the algorithm's capacity to discover stable physiology–emotion interactions despite heterogeneous experimental protocols. Figure **??** visualizes shared rule generalization. In heatmap (c) of Figure **??**, the pairwise overlapping between datasets is depicted. The highest overlapping occurs between CASE and EmoWorker (306 shared rules) and between CASE and K-emoCon (ext) (295 shared rules), while EMBOA shows the least overlapping with other datasets (7–43 shared rules); still, this dataset is for autistic children–a fact possibly influencing the obtained result, yet not incorporated as a variable in the research. The next three heatmaps ((a), (b), (d)) show, respectively, pairwise generalization (persistence in a third dataset), triplet generalization (persistence in a fourth dataset), and quadruplet analysis (persistence in a fifth dataset). The results of quintuplet analysis are not presented as no common rules were obtained. Based on the premise that the validity of a pattern increases with its prevalence across multiple independent datasets, and applying the 'majority rule' (consensus in at least four out of six datasets), 1+1+2+54 rules were identified as potentially generalized patterns (depending on the datasets, in which they are approved). However, it is important to exercise caution in this classification; since two of the confirming datasets were derived from the same dataset (K-emoCon), the assumption of total independence is not fully met for the 54 rules. Despite this constraint, these rules still demonstrate a high level of credibility. Furthermore, a significant number of rules were confirmed across three datasets. These are categorized as domain-robust patterns, as they exhibit substantial resilience to environmental changes and suggest strong reproducibility within similar experimental contexts.

To assess the influence of annotation source (**RQ1.1**), the union of all rules from the four self-annotated datasets (7095 unique rules) was compared with the union from the two externally annotated datasets (4598 unique rules). A Jaccard similarity of 0.038 (432 shared rules) reveals a relatively small overlap, suggesting that while a core set of physiology–emotion patterns exists, the vast majority of findings are heavily influenced by the annotation paradigm.

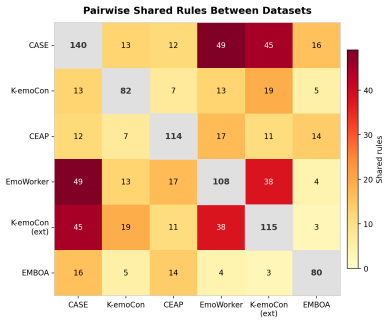
As part of Experiment 1, regarding **RQ1.2**, the rules extracted for the dimensional model and the discret model were compared (see Table **??**) across all six datasets. The rules were filtered analogically to the dimensional model, i.e., only cross-modal rules linking an emotional component to a physiological signal were retained. The discrete model produced fewer cross-modal rules in every dataset, yielding only 12–40% of the dimensional model's rule count. The largest reduction was observed for self-annotated K-emoCon (144 vs. 1208 rules; ratio = 0.12), whereas the smallest gap appeared in EMBOA (1184 vs. 2972; ratio = 0.40). Across all six datasets, the

figures/heatmap_pairs_generalization_sparse.png

(a) Pairwise Generalization

figures/heatmap_triplets_generalization_sp

(b) Triplet Generalization



(c) Pairwise Shared Rules

figures/heatmap_quadruplets_generalization_sparse.pn

(d) Quadruplet Generalization

Fig. 1. Shared rule generalization

dimensional model produced 12,206 filtered rules, while the discrete model produced 2,951, corresponding to an overall ratio of 0.24. These results indicate that the dimensional representation preserves a substantially richer set of cross-modal emotion–physiology patterns, whereas the discrete representation leads to a stronger reduction in retained rules and a sparser cross-modal structure. Intersection analyses for a discrete model do not provide the identification of generalized patterns. Approximately 30 domain-robust patterns were identified across three datasets as depicted in Figure **??**.

Experiment 2. Within this experiment the rules qualified in Experiment 1 as generalized patterns are analysed. The first observation is *the predominance of moderate affective states and HRV synchronicity*. The discovery process revealed a significant density of rules converging on the `valence_medium` and `arousal_medium` states. A recurring motif in the find-

figures/heatmap_discrete_pairs_generalization_sparse.p

figures/heatmap_discrete_pairwise_sparse.p

(a) Discrete Pairwise Generalization

(b) Discrete Pairwise Shared Rules

figures/heatmap_discrete_triplets_generalization_sparse.png

(c) Discrete Triplet Generalization

Fig. 2. Shared rule generalization (Discrete model)

ings is the high degree of synchronicity between multiple HRV indices. Specifically, the frequent occurrence of the equals predicate among features such as `hrv_rmssd`, `hrv_sdnn`, and `hrv_cvnn` indicates that these parameters do not act in isolation. Instead, they form a "physiological cluster" where their simultaneous stabilization—whether at low or high levels—serves as a robust precursor to a moderate affective state. This suggests that the model favors a holistic physiological signature over individual feature fluctuations. The second observation is related to *physiological lag and the `meets` transition logic*. A critical finding of this study is the consistent appearance of the `meets` predicate in rules linking physiological changes to emotional responses. In the context of the ARMADA algorithm, this indicates a zero-gap temporal transition, where the end of a specific heart rate variability or skin conductance interval coincides exactly with the onset of an affective state classification. This is particularly evident in the rules involving `hrv_cvsd_low` and `hrv_rmssd_low`. These patterns effectively model the inherent physiological lag of the autonomic nervous system, suggesting that the "moderate" emotional response is a direct consequence of a preceding period of reduced cardiac variability, likely associated with cognitive load or emotional processing. A third notable finding in the discovery process is *the unique temporal association between low-arousal states and*

electrodermal activity, as captured by the rule: `arousal_low` → `arousal_low meets eda_scr_auc_medium`. Unlike the HRV-based rules, which primarily characterize moderate affective states, this specific rule highlights a distinct physiological marker for the termination of low-arousal phases. The application of the `meets` predicate suggests that the conclusion of a low-prowess state is immediately followed by a moderate increase in the Area Under the Curve (AUC) of skin conductance responses (SCR). This indicates that while HRV metrics are robust predictors of sustained moderate arousal, EDA phase-activity features are more sensitive to the transitions occurring at the lower end of the arousal spectrum.

When regarding RQ3 the analysis is again focused on the generalized patterns from Experiment 1. It is interesting that within these rules there are only 3 concerning EDA and 48 for HRV. There are no rules for the temperature. Interestingly, the rule discovery process did not yield any multimodal rules combining HRV and EDA features within a single antecedent. This suggests that in the study, cardiac and electrodermal responses manifest as independent temporal events rather than synchronized physiological patterns, manifesting modality independence.

10. Conclusions

The efficacy of the ARMADA algorithm in extracting human-readable, temporal association rules linking physiological signals to emotional transitions was demonstrated.

Despite the inherent variability across the different datasets, over 50 specific rules met the rigorous criteria for cross-dataset validation, demonstrating their consistency and reliability. It was found that rule quality is significantly impacted by the choice of experiment methodology that led to dataset creation. A substantially richer set of patterns was provided by the multidimensional valence-arousal space compared to discrete categories, with the latter yielding only 12–40% of the former's rule count. Furthermore, it was observed through a low Jaccard similarity (0.038) that the source of emotional labels—specifically self-reports versus expert external annotations—leads to distinct sets of discovered temporal rules, suggesting that physiological manifestations are highly sensitive to the annotation paradigm.

The importance of modeling physiological lag was highlighted through the discovery of patterns utilizing the `meets` predicate. Specific HRV-based clusters were identified as robust temporal precursors to moderate affective states. Conversely, transitions at the lower end of the arousal spectrum were found to be uniquely associated with electrodermal activity (EDA).

Contrary to expectations, multi-modal association rules were not found to be dominant over single-modal rules. It was observed that cardiac and electrodermal responses often manifest as independent temporal events rather than synchronized patterns.

A foundation for building transparent, rule-based classifiers was offered by the identified generalized patterns (universal across datasets) and domain-robust patterns (resilient across specific contexts).